Chapter 5

1-9 and 11

1. To iterate code efficiently without having to write the same code multiple times
2. While loop checks condition before loop, do while loop checks condition after running once (at the end)
3. Review: Account setup
4. a) A loop that doesn't end because the condition to stop is not fulfilled b) While True loop that doesn't change to false or while it is an integer the counter will never reach c) when a calculation exceeds the maximum limit of a data type like an integer or stack memory
5. The loop will run 60 times
6. Any number greater than or equal to 123 (can't be 120 as for do-while loop first iteration happens no matter what)
7. A counter is mainly used to count the amount of iterations or loops and is incremented by a constant (not exponential). An accumulator is to calculate a group of volumes like a sum and can be increased by any amount.
8. int sum = 0; // Initialize sum at 0
9. for (int i = 3; i <= 10; i++) {sum += i;} System.out.println(sum);

11. a) x.length = 9 b) x.substring(0, 3) [my]

Chapter 6

2-9, 11

2. Declaration is how to address the method (public/private static, returns/void) this (the name) is what is called in the main method. The method body is where the actual code and function is written in. It is enclosed in curly braces

3. The access modifier

4. Visibility

5. Var3: local to function not loop, Var 4: local to function and in for loop, Var1: Local to main not loop, Var2: Local to main and loop

6. a) Public static int getVowels(String str) b) public static int extractDigit(int dig) c) public static String insertString(String str, int in)

7. a) from the parameters of the method and its name b) two methods can have the same name and class but they must differ in at least one parameter or order of parameters

8. a) returns a value from the function b) only one value c) one parameter will show a value in the declaration for type, the other will show void for no value

9. Extra bracket that ends class too fast

11. a) true b) false, a method call calls an instance of the method declaration not the actual declaration c) false it must not d) false an access modifier determines what other classes/methods can call it e) true f) true g) false local variables can only be used from the method it is in h) i) false method overloading is 2 different methods with the same name and different parameters j) true k) false, precondition of a method states requirements for the

function to work that may not be in the parameter l) false, postcondition only shows the final result not how it gets there.